CDC PLACES Data Exploration

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1. Introduction

Chronic diseases such as asthma and arthritis are major public health concerns in the United States. These conditions may co-occur and share common environmental, behavioral, and demographic risk factors.

This project uses the CDC PLACES 2023 county-level dataset to examine the relationship between asthma prevalence and arthritis prevalence across U.S. counties.

Research Question:

Is there an association between asthma prevalence and arthritis prevalence across U.S. counties?

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2. Materials & Methods

Data Source

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)

places <- read.csv("~/Desktop/6270 assignment/CP02/PLACES__Local_Data_for_Better_Health__County_Data_20
```

Variables

Asthma prevalence: percentage of adults aged >=18 with current asthma Arthritis prevalence: percentage of adults aged >=18 with arthritis

Methods

The analysis includes:

Descriptive statistics (mean, median) Data visualization (histogram and scatterplot) Correlation analysis

3. Results

3.1 Asthma Prevalence

```
asthma <- places %>%
  filter(MeasureId == "CASTHMA") %>%
  select(StateDesc, LocationName, Data_Value) %>%
  rename(Asthma = Data_Value)

mean(asthma$Asthma, na.rm = TRUE)
```

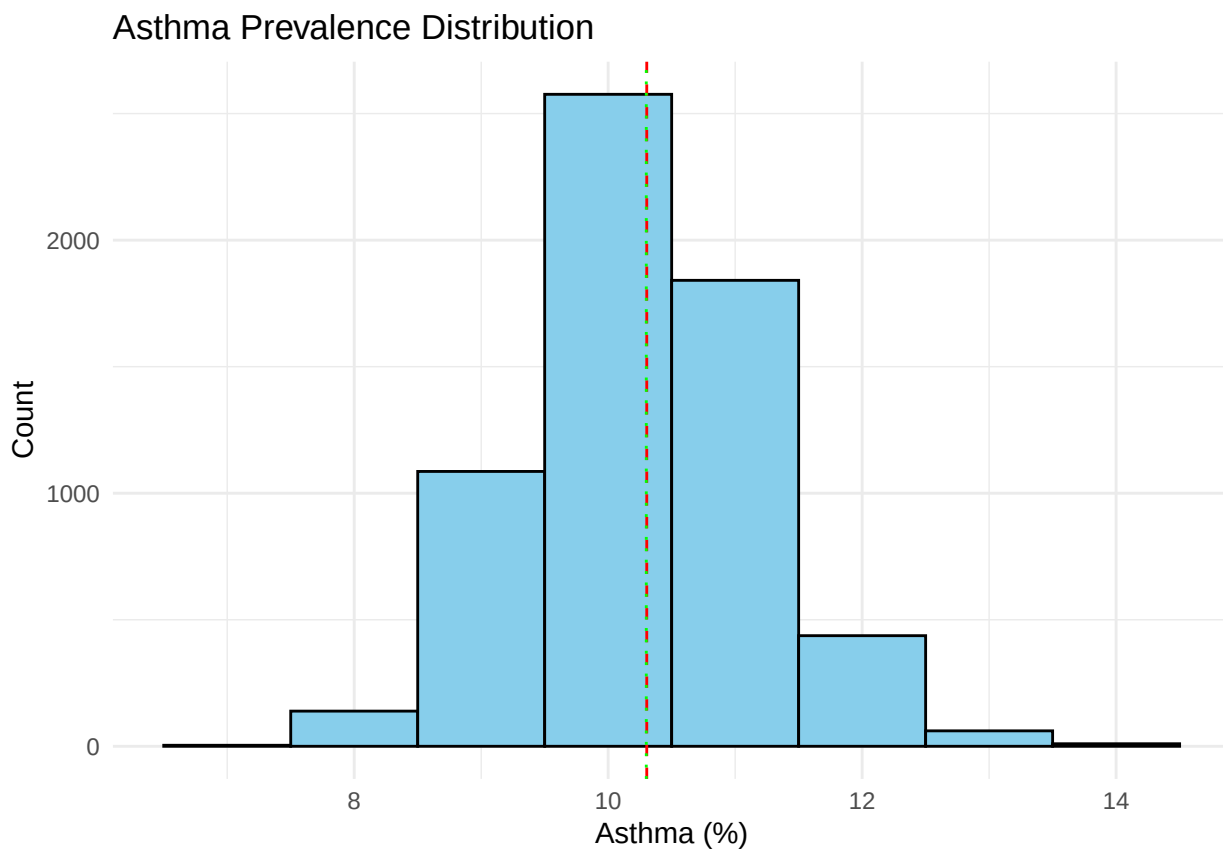
```
## [1] 10.3039
```

```
median(asthma$Asthma, na.rm = TRUE)
```

```
## [1] 10.3
```

Distribution of Asthma

```
ggplot(asthma, aes(x = Asthma)) +
  geom_histogram(binwidth = 1, fill = "skyblue", color = "black") +
  geom_vline(xintercept = mean(asthma$Asthma, na.rm = TRUE),
             color = "red", linetype = "dashed") +
  geom_vline(xintercept = median(asthma$Asthma, na.rm = TRUE),
             color = "green", linetype = "dotted") +
  labs(title = "Asthma Prevalence Distribution",
       x = "Asthma (%)",
       y = "Count") +
  theme_minimal()
```



3.2 Relationship Between Asthma and Arthritis

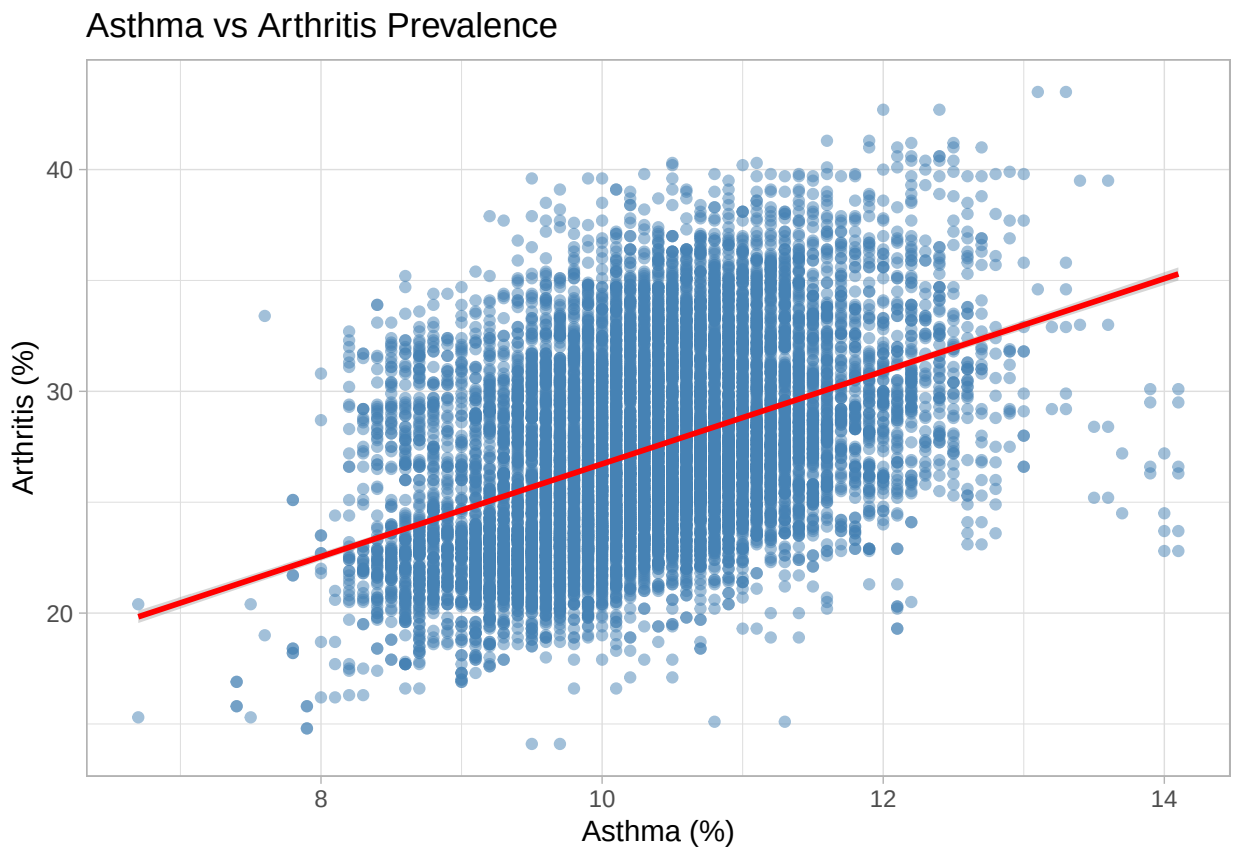
```
arthritis <- places %>%
  filter(MeasureId == "ARTHRITIS") %>%
  select(StateDesc, LocationName, Data_Value) %>%
  rename(Arthritis = Data_Value)

df <- left_join(asthma, arthritis,
                by = c("StateDesc", "LocationName")) %>%
  na.omit()
```

```
## Warning in left_join(asthma, arthritis, by = c("StateDesc", "LocationName")): Detected an unexpected
## i Row 1 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 3158 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

Scatterplot

```
ggplot(df, aes(x = Asthma, y = Arthritis)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(title = "Asthma vs Arthritis Prevalence",
       x = "Asthma (%)",
       y = "Arthritis (%)") +
  theme_light()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Correlation

```
cor(df$Asthma, df$Arthritis, use = "complete.obs")
```

```
## [1] 0.4380106
```

The correlation (r is approximately 0.44) indicates a moderate positive association between asthma and arthritis prevalence across U.S. counties.

4. Conclusion

This study examined the relationship between asthma and arthritis prevalence using CDC PLACES county-level data. The results indicate a moderate positive association (r = 0.44), suggesting that counties with higher asthma prevalence also tend to report higher arthritis prevalence.

This pattern may reflect shared underlying risk factors such as socioeconomic conditions, environmental exposures, and health behaviors. However, this is an observational ecological analysis, so no causal inference can be made.

Limitations include:

Cross-sectional design No control for confounding variables Ecological nature of the data

5. Interactive Shiny Application

An interactive Shiny application was developed to complement this analysis. The application allows users to explore distributions and relationships between variables dynamically.

The app includes histogram visualizations, a scatterplot with a regression line, and summary statistics, which together support exploratory data analysis and improve interpretability of relationships between health indicators.

The Shiny application is available at: https://ruitingzhu.shinyapps.io/my_app/

AI Disclosure

ChatGPT was used to assist with structuring the R Markdown report, improving code syntax, and refining written interpretation. All analyses and results were generated and verified by the author.